Semisupervised Building Instance Extraction From High-Resolution Remote Sensing Imagery

Fang Fang, *Member, IEEE*, Rui Xu, Shengwen Li, *Member, IEEE*, Qingyi Hao, Kang Zheng, Kaishun Wu, and Bo Wan, *Member, IEEE*

***Abstract*— Automatic building instance extraction from high-resolution (HR) remote sensing imagery (RSI) is crucial for urban planning and mapping. The dominant approaches are based on the full-supervised learning paradigm that requires a large number of labeled samples to train their models, which is very time-consuming and labor-intensive. To alleviate this problem, this study proposes a semisupervised building instance extraction method that integrates teacher–student learning and pseudo-labeling to improve the building instance extraction from HR RSI. Specifically, the proposed method consists of three modules, the hybrid data augmentation (HDA) module, the pseudo-label generation (PLG) module, and the contour refinement (CR) module. The HDA module is designed to enrich the diversity of labeled samples to optimize the teacher model. The PLG module generates pseudo-labels from unlabeled data to train a student model with pseudo-labels. Finally, the CR module is designed to refine the contours of buildings. Experimental results on three challenging public datasets demonstrate that the proposed method achieves superior performance and exhibits great robustness at different proportions of labeled data and different building scenarios. This study provides a new approach for extracting building instances from HR RSI in scenarios with insufficient labeled samples and a methodological reference for various applications of semisupervised on RSIs.**

***Index Terms*— Building instance extraction, high-resolution (HR) remote sensing imagery (RSI), pseudo-label, semisupervised.**

I. Introduction

AUTOMATIC building extraction from remote sensing imagery (RSI) has been widely deployed in urban planning, environmental monitoring and management, photogrammetry, and map updating [1], [2], [3]. With the blossoming of Earth observation technology, high-resolution (HR) RSIs have become readily available, providing great support for timely and accurate building extraction, while HR RSIs bring a lot of redundant information that is not related to building instance features. Accurately extracting buildings from HR RSI is still very challenging.

In the last decade, many efforts have been made to extract buildings from RSI. Traditional building extraction methods rely on the handcrafted building features, such as edge, shadow, color, shape, and so on [4], [5]. Since building features vary dramatically in different regions, building features need to be redesigned for each region, which requires extensive expert experience and is difficult to be replicated to other regions.

Recently, deep learning-based methods have shown superiority in building extraction by automatically learning building features [1], [3], [6], [7]. They consider building extraction as a semantic segmentation task, in which buildings are extracted by classifying RSI pixels as buildings or nonbuildings. For example, full convolutional networks (FCNs) [8], which are widely used in semantic segmentation tasks, were introduced into building extraction and further combined with machine learning techniques, such as conditional random fields to enhance building extraction [3]. Based on the U-Net [9] network, a Siamese U-Net with shared weights in two branches is proposed to achieve better segmentation accuracy [1]. Although these methods can classify an RSI into building and nonbuilding areas, they cannot extract individual building instances.

The building instance extraction task aims at extracting individual buildings from RSIs, which can be considered as a kind of instance segmentation task [10]. Represented by Mask R-CNN [11], some instance segmentation-based building extraction methods have been proposed and made great progress in recent years [2], [12], [13], [14], [15]. For example, Xu et al. [2] introduced gated spatial memory module and centroid-aware head into Mask R-CNN to promote accurate and complete recognition for irregularly shaped buildings. Fang et al. [14] proposed a coarse-to-fine contour optimization network to improve the ability to detect small buildings and refine building contours. These methods all follow the full-supervised machine learning paradigm, which requires massive pixel-level labeled samples to train their models. In practice, buildings in HR RSIs are highly complex

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Fang Fang is with the School of Computer Science and the National Engineering Research Center of Geographic Information System, China University of Geosciences, Wuhan 430074, China, and also with the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China (e-mail: fangfang@cug.edu.cn).

Rui Xu is with the School of Computer Science, China University of Geosciences, Wuhan 430074, China, and also with the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China (e-mail: uiirux@cug.edu.cn).

Shengwen Li and Bo Wan are with the School of Computer Science and the National Engineering Research Center of Geographic Information System, China University of Geosciences, Wuhan 430074, China (e-mail: swli@cug.edu.cn; wanbo@cug.edu.cn).

Qingyi Hao, Kang Zheng, and Kaishun Wu are with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: 1202110795@cug.edu.cn; zhengkang@cug.edu.cn; ks@cug.edu.cn).

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and diverse, and labeling samples is very labor-intensive and time-consuming.

To alleviate the dependence of model training on labeled samples, this article proposes a semisupervised building instance extraction method that extracts individual buildings from HR RSIs by leveraging limited labeled data and abundant unlabeled data. Specifically, a hybrid data augmentation (HDA) module is devised to increase the diversity of labeled training data. Also, a pseudo-label generation (PLG) module is derived from the teacher–student network to generate usable pseudo-labels for unlabeled data. Finally, a contour refinement (CR) module is designed to alleviate the negative effects of inaccurate pseudo-labels and, thus, refine the extracted contours. To the best of our knowledge, this is the first attempt to introduce semisupervised learning (SSL) into building instance extraction from HR RSIs. The main contributions of this article can be summarized as follows.

1) This study highlights the necessity of the semisupervised paradigm in instance extraction for HR RSI and proposes to introduce semisupervised techniques to improve building instance extraction from HR RSIs.
2) This study develops a semisupervised building instance extraction framework, which integrates teacher–student learning and pseudo-labeling for building instance extraction. In addition, the framework designs an HDA module to increase both the size and diversity of labeled data, and a CR module to refine building contours.
3) Experiments on three datasets demonstrate that the proposed method enhances the performance of building instance extraction and outperforms the state-of-the-art fully supervised and semisupervised instance segmentation methods.

This study is organized as follows. Section II provides a brief review of related works. Section III describes the proposed method in detail. The experimental results are reported and analyzed in Section IV. Section V discusses ablation studies and some factors that may impact model performance and robustness. Finally, Section VI concludes this study.

II. Related Work

Related works can be classified into two categories, building extraction based on instance segmentation and semisupervised building extraction from RSIs. They are discussed in Sections II-A and II-B.

A. Building Extraction Based on Instance Segmentation

The instance segmentation methods fall into two groups, one-stage methods and two-stage methods. The one-stage methods [16], [17], [18], [19], such as SOLO [18] and SOLOv2 [19], follow the process of performing pixel-level segmentation and grouping the pixels into different instances. Several one-stage-based methods have been proposed to extract building instances. Wagner et al. [20] presented an architecture that consists of three parallel paths for the instance segmentation of buildings in very HR satellite imagery. TernausNetV2 [21] proposed an encoder–decoder network with skipped connections that allows extracting objects from HR satellite imagery on an instance level.

The two-stage methods [11], [22], [23], [24], [25], [26] first perform object detection and then predict segmentation masks, including Mask R-CNN [11], Cascade R-CNN [23], and RefineMask [26]. These methods have made notable advancements in building instance extraction. Zhao et al. [12] presented a method that combines Mask R-CNN with building boundary regularization to produce better-regularized polygons. Liu et al. [13] designed a multiscale U-shaped CNN building instance extraction framework with edge constraints to extract precise building masks and suppress overfitting. Wang et al. [15] integrated path aggregate feature pyramid network (FPN) and atlas space pyramid pool into Mask R-CNN to improve the multiscale feature extraction and fusion capability of traditional village building extraction.

However, all these methods require massive pixel-level and bounding box annotated samples for model training, which is time-consuming and labor-intensive.

B. Semisupervised Building Extraction From RSIs

SSL, which uses both labeled and unlabeled data to train models, has emerged as a powerful paradigm that mitigates the dependence on large labeled datasets [27]. Previous work has developed a range of SSL techniques, including consistency regularization [28], [29], [30], pseudo-labeling [31], [32], [33], [34], data augmentation [35], [36], [37], and holistic methods [27], [37], [38].

For RSI building extraction tasks, a growing literature has proposed various semisupervised approaches. For example, Xia et al. [39] proposed a semantic edge detection approach for very-HR RSIs based on SSL and modified D-LinkNet to improve the quality of pseudo-labels. Li et al. [40] proposed a semisupervised building footprint generation method based on semantic segmentation, which alleviates the limitation of insufficient training samples with annotations by consistent training of features and outputs. Guo et al. [41] introduced a saliency-guided edge-preservation network (SG-EPUNet) that maintains both localization accuracy and edge precision to generate up-to-date building footprints.

The above methods have advanced building extraction. These works focus on building extraction with the semantic segmentation methods, aiming to separate RSI into building and nonbuilding areas. That is, they cannot be used to extract building instances.

III. Methods

A. Overview

This study proposes to introduce SSL for building instance extraction from HR RSIs. In this article, a set of HR RSIs with building labels and without building labels are denoted as $D_s = \{x_i^s, y_i^s\}_{i=1}^{N_s}$ and $D_u = \{x_i^u\}_{i=1}^{N_u}$, respectively, where N_s and N_u denote the number of labeled and unlabeled images, respectively, and $N_u \gg N_s$. For each labeled image x_i^s, the annotation y_i^s mainly consists of instance category c_i^s (only building), locations of all bounding boxes r_i^s, and masks m_i^s with the building instance id. The building instance extraction of SSL is to accurately extract building instances from RSIs by jointly exploiting both D_s and D_u.

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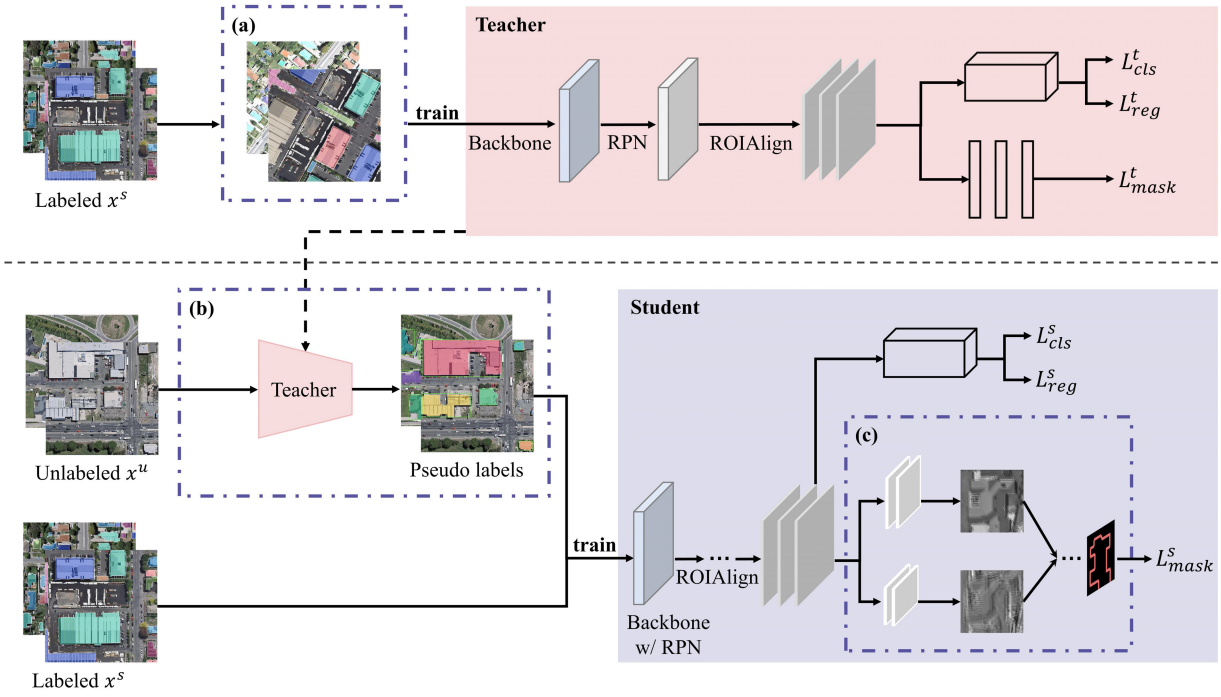


Fig. 1. Overview of the proposed method. (a) HDA. (b) PLG. (c) CR.

As shown in Fig. 1, the proposed method is derived from the classical teacher–student network and consists of three modules, namely, HDA, PLG, and CR. The HDA module designs an HDA strategy to assist the teacher model learning on labeled data. The PLG module uses the teacher model as a detector and generates pseudo-labels for the unlabeled data. The CR module is designed to help the student model refine the extracted building contours.

B. HDA Module

The HDA module is designed to enrich the building samples and features, which consists of two steps, the geometric data augmentation strategy and the basic data augmentation strategy.

The geometric data augmentation strategy first selects a point in RSI as its center and then performs geometric transformation operations on both the RSI and its ground-truth (GT) image. The operations include random angle rotation, scaling up/down to random factors, and filling blank areas by reflecting. An image transformed by the strategy is illustrated in Fig. 2. As shown in the figure, there are several incomplete buildings in the original RSI appearing on the edges of tiles, which is mainly caused by the RSI slicing. With the geometric transformation augmentation, these truncated buildings can be complemented.

After geometric data augmentation, the RSIs are enhanced with the basic data augmentation strategy, which consists of a set of operations, including scale jitter, random flip, sharpness jitter, and brightness jitter. These operations are performed sequentially with certain probabilities.

C. PLG Module

The PLG module is designed to generate pseudo-labels of unlabeled RSIs. Specifically, the building instance labels

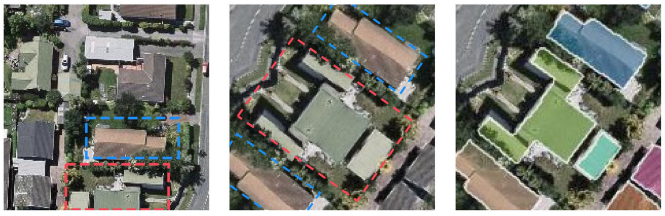


Fig. 2. Example of geometric data augmentation. (a) Original image. (b) Image with geometric data augmentation. (c) GT of the image after augmentation. Note: the red boxes represent the incomplete buildings located in the edge of the tiles, and the blue boxes represent the ordinary intact buildings in tiles.

are generated from unlabeled RSIs by the pretrained teacher network. Following the work [42], the module uses scaling jitter and horizontal flip to improve the label quality.

To obtain reliable pseudo-labels, this module uses a heuristic adaptive threshold strategy to filter the false positive pseudo-labels. Here, we assume that the unlabeled and labeled RSIs follow similar distribution of buildings. Specifically, the average number of building instances in each labeled image is calculated as follows:

$$\theta^s = \frac{1}{N_s}\sum_i k_i^s \tag{1}$$

where k_i^s denotes the number of building instances in the ith labeled RSI.

Then, the average number of predicted building instances can be calculated as follows:

$$f(\mu_j) = \left\{ \frac{1}{N_u}\sum_i \left(\ell_{\{c_i^u > \mu_j\}} k_i^u \right) \right\} \tag{2}$$

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Algorithm 1 Generating Pseudo-Labels

Input: unlabeled images x_i^u, step τ
Output: pseudo labels $\hat{y}_i^u$
Generate the random augmented images:
$a(x_i^u) = augmentation(x_i^u)$
Forward x_i^u through teacher model to generate prediction:
$(c_i^u, r_i^u, m_i^u) = teacher(a(x_i^u))$
Calculate the average of instances for labeled image by Equation 1
Set $j = 0$, the initial threshold $\mu_j = 0.1$
while $\mu_j \leq 1$ **do**
Calculate the averages of predictions by Equation 2
$\mu_{j+1} = \mu_j + \tau$
$j = j + 1$
end while
Calculate threshold μ' by Equation 3
Filter the prediction $\hat{y}_i^u$ by Equation 4
return $\hat{y}_i^u$

where k_i^u is the number of filtered building instances in the ith unlabeled RSI. c_i^u is the predicted score of the ith unlabeled RSI. μ_j is the filtering threshold.

The threshold μ' is determined by the difference between θ_j^u and θ^s as follows:

$$\mu' = \arg\min_{\mu_j} \left(|f(\mu_j) - \theta^s| \right). \tag{3}$$

Finally, the pseudo-labels are predicted as follows:

$$\hat{y}_i^u = \left\{ \ell_{\{c_i^u > \mu'\}} \left(c_i^u, r_i^u, m_i^u \right) \right\} \tag{4}$$

where r_i^u and m_i^u are the predicted bounding boxes and building instance masks of the ith unlabeled RSI, respectively.

The whole process of generating pseudo-labels is described by Algorithm 1.

D. CR Module

Although the PLG module has filtered out some false positive pseudo-labels, there is still more or less noise in pseudo-labels. This noise is usually near building contours, because the pixels at this location are at the decision boundary and have less discriminative features [43]. As shown in Fig. 3, there is some noise in the pseudo-labels near the building contours.

The CR module is to reduce the effect of pseudo-label contour noise and improve building extraction contours. The structure of the CR module is illustrated in Fig. 4. Specifically, the features of the mask branch are fed into two scale branches, a large-scale branch of 28×28 size and a small-scale branch of 14×14 size, respectively. The features in the former branch help to capture more detailed features of the building. Also, the features in the latter branch help to highlight the whole of the building and reduce the effect of local noise. Fusing the features of the two branches will obtain the overall building information while preserving building details.

Although the two branches alleviate the effect of contour noises from pseudo-labels, the deviation between the extracted building contours and the GTs still exists. To further refine the building instance contours, a two-channel Sobel operator [44] is used to identify building contours. Specifically, the Sobel operator is used to locate building contours in the horizontal and vertical directions and integrates the output features from the two channels to further refine building contours. Once the building contours are determined, the building instances can be extracted.



Fig. 3. Illustration of building contour noise in pseudo-labels. The white circles represent building contours with noise.

E. Model Training

1) Teacher Model Training: The teacher model, a Mask R-CNN network, is trained with the labeled data. The optimization loss L^t of this model consists of five parts, including the region proposal network (RPN) classification loss $L_{\text{cls}}^{\text{rpn}}$, the RPN regression loss $L_{\text{reg}}^{\text{rpn}}$, the region of interest (ROI) classification loss $L_{\text{cls}}^{\text{roi}}$, the ROI regression loss $L_{\text{reg}}^{\text{roi}}$, and the mask loss L_{mask} [11]. The total loss L^t is formulated as follows:

$$\begin{aligned} L^t = \sum_i & L_{\text{cls}}^{\text{rpn}}\left(x_i^s, y_i^s\right) + L_{\text{reg}}^{\text{rpn}}\left(x_i^s, y_i^s\right) + L_{\text{cls}}^{\text{roi}}\left(x_i^s, y_i^s\right) \\ & + L_{\text{reg}}^{\text{roi}}\left(x_i^s, y_i^s\right) + L_{\text{mask}}\left(x_i^s, y_i^s\right). \end{aligned} \tag{5}$$

2) Student Model Training: In this study, pseudo-labeled data and labeled data are used to train the student model. In the process of the student model training, the basic data augmentation in Section III-B is extended by adding Gaussian noise, Gaussian blur, and invert transformations. The geometric data augmentation is not performed to reduce the impact of noise from the pseudo-labels. The loss function of unlabeled data is formulated as follows:

$$\begin{aligned} L_{\text{unsup}}^s = \sum_i & L_{\text{cls}}^{\text{rpn}}\left(x_i^u, \hat{y}_i^u\right) + L_{\text{reg}}^{\text{rpn}}\left(x_i^u, \hat{y}_i^u\right) + L_{\text{cls}}^{\text{roi}}\left(x_i^u, \hat{y}_i^u\right) \\ & + L_{\text{reg}}^{\text{roi}}\left(x_i^u, \hat{y}_i^u\right) + L_{\text{mask}}\left(x_i^u, \hat{y}_i^u\right) \end{aligned} \tag{6}$$

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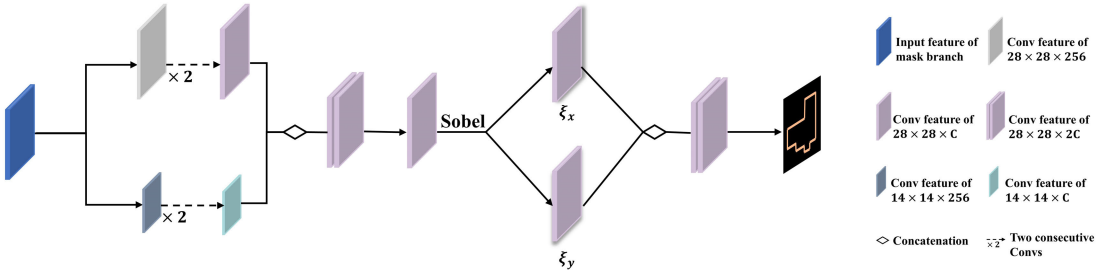


Fig. 4. Structure of the CR module. In the figure, ξ_x represents the horizontal direction feature, and ξ_y represents the vertical direction feature.



Fig. 5. Samples of the WHU aerial building dataset.



Fig. 6. Samples of the CrowdAI mapping challenge dataset.

where $\{\hat{y}_i^u\}_{i=1}^{N_u}$ is the pseudo-label of the unlabeled data. Also, the loss of the labeled data is consistent with (5)

$$L_{\text{sup}}^s = L^t. \tag{7}$$

By combining (6) and (7), the total loss of the student model is

$$L^s = L_{\text{unsup}}^s + L_{\text{sup}}^s. \tag{8}$$

IV. Experiments and Results

A. Datasets

The proposed method was evaluated on three public building extraction datasets, the WHU aerial image dataset [1], the CrowdAI mapping challenge dataset [45], and the building dataset of Typical Cities in China (TCC) [46].

1) WHU Aerial Building Dataset: The WHU aerial building dataset covers an area of approximately 450 km^2 and contains over 187 000 buildings of various sizes and appearances in Christchurch, New Zealand. It consists of 8188 RGB images with 512 × 512 pixels at a 0.3-m spatial resolution. The dataset is divided into three sets: 4736 images for training, 1036 images for validation, and 2416 images for testing. Both the aerial images and their corresponding GT images are included in the dataset. Fig. 5 shows some examples from the dataset, where building instances are annotated with colored GT labels.

2) CrowdAI Mapping Challenge Dataset: The CrowdAI mapping challenge dataset provides over 340 000 individual satellite images tiles of 300 × 300 pixel RGB images with a spatial resolution of 0.3 m and building annotations. The training dataset consists of over 280 000 satellite images, and the test dataset contains over 60 000 images. The images are available in JPEG format, along with their annotations in the MS COCO format. Fig. 6 shows some examples from the CrowdAI dataset.

3) TCC Building Dataset: The TCC building dataset consists of 7260 tiles and 63 886 building instances in four Chinese cities: Beijing, Shanghai, Shenzhen, and Wuhan. The training and test sets consist of 5985 and 1275 images, respectively. Each image has a size of 500 × 500 pixels with a resolution of 0.29 m. Some samples from the dataset are shown in Fig. 7. The dataset includes nonorthophotos images, which are challenging for building instance extraction tasks.

B. Experimental Settings

We use partial data as labeled data to verify the performance of the model in semisupervised scenarios. Specifically, 5%,

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TABLE I
COMPARISON RESULTS ON THREE DATASETS

Dataset	Method	mAP					
		5%	10%	20%	30%	40%	100%
WHU	Mask R-CNN	0.514	0.563	0.589	0.600	0.614	0.635
	RefineMask	0.547	0.591	0.619	0.634	0.641	**0.653**
	SOLOv2	0.451	0.527	0.566	0.589	0.608	0.621
	NB	0.559	0.608	0.631	0.643	0.660	-
	Proposed method	**0.582**	**0.625**	**0.644**	**0.658**	**0.670**	-
CrowdAI	Mask R-CNN	0.404	0.466	0.507	0.529	0.543	0.562
	RefineMask	0.482	0.521	0.546	0.558	0.568	**0.585**
	SOLOv2	0.412	0.475	0.508	0.518	0.546	0.569
	NB	0.467	0.511	0.542	0.554	0.564	-
	Proposed method	**0.506**	**0.529**	**0.551**	**0.564**	**0.573**	-
TCC	Mask R-CNN	0.215	0.291	0.366	0.385	0.409	0.425
	RefineMask	0.295	0.362	0.411	0.430	0.441	**0.459**
	SOLOv2	0.171	0.289	0.368	0.380	0.401	0.420
	NB	0.298	0.355	0.402	0.425	0.438	-
	Proposed method	**0.320**	**0.372**	**0.414**	**0.433**	**0.445**	-



Fig. 7. Samples of the TCC building dataset.

10%, 20%, 30%, and 40% images from the training set are sampled as labeled training data, and the remaining images are used as unlabeled data. All experiments are performed with five data folds, and the final performance is calculated as the average of all five folds [43], [47].

The basic networks of the teacher model and student model adopt the Mask R-CNN [11] with FPN [48] and use ImageNet pretrained ResNet-50 [49] as their backbone. All experiments are implemented in PyTorch [50] and MMDetection [51] and run on a 3090 GPU. The stochastic gradient descent (SGD) training [52] is employed and initialized with a learning rate of 0.0025. The weight decay and the momentum are set to 0.0001 and 0.9, respectively.

C. Evaluation Metrics

In this study, the widely used mean average precision (mAP) of the standard MS COCO metrics [53] is employed for evaluating the task. The mAP is computed at ten intersection-over-union (IoU) overlap thresholds ranging from 0.50 to 0.95 in 0.05 increments, as follows:

$$\text{mAP} = \frac{\text{AP}_{0.50} + \text{AP}_{0.55} + \cdots + \text{AP}_{0.90} + \text{AP}_{0.95}}{10}. \quad (9)$$

D. Baselines

To investigate the performance of the proposed method, four state-of-the-art methods are selected as baselines. The baselines include three supervised building instance extraction methods, Mask R-CNN [11], RefineMask [26] and SOLOv2 [19], and a semisupervised instance segmentation method, noisy boundaries (NBs) [43]. Mask R-CNN is the basic network of the proposed method and is widely used to extract building instances. RefineMask, an excellent two-stage instance segmentation method, incorporates fine-grained features during the instance-wise segmentation process in a multistage manner, which consistently refines high-quality masks through fusing more detailed information stage by stage. SOLOv2 is a typical representative of one-stage building instance extraction method, which can dynamically segment each instance in an image without resorting to bounding box detection and is used as a baseline in many studies [54], [55]. NB is the first semisupervised instance segmentation framework in computer vision field. All the baseline methods adopted the default settings of the MMDetection framework [51].

E. Overall Results

The experimental results on the WHU, CrowdAI, and TCC datasets are reported in Table I. The mAP values with different ratios of labeled data are listed in the table. Among them, the

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mAP of 100% labeled ratio means the result of full-supervised methods. As shown in Table I, the proposed method achieves the best results at different ratios of labeled data on all three datasets.

For the WHU dataset, the proposed method outperforms RefineMask, the best of all supervised baselines, by 3.5%, 3.4%, 2.5%, 2.4%, and 2.9% in terms of mAP when the labeled ratio is 5%, 10%, 20%, 30%, and 40%, respectively. Furthermore, the proposed method requires only 30% of the labeled data to obtain the value of 65.8% in mAP, which outperforms RefineMask, SOLOv2, and Mask R-CNN with 100% of the labeled data. Compared with NB, the state-of-the-art semisupervised instance segmentation method, the proposed method is superior at all labeled ratios. For example, the mAP improves by 1.7% and 1.5% when the labeled ratios are 10% and 30%, respectively.

For the CrowdAI dataset, the proposed method still achieves the highest mAP in different proportions of labeled data. In particular, compared with Mask R-CNN, SOLOv2, RefineMask, and NB, the proposed method outperforms 10.2%, 9.4%, 2.4%, and 3.9% in terms of mAP when the labeled ratio is 5%. These results show that our approach has great potential in dealing with scenarios with insufficient labeled samples.

The TCC dataset contains nonorthophoto images with complex backgrounds, which makes the extraction of building instances very challenging. The mAP of all models is lower than those of the above two datasets. The proposed method achieves higher mAP than RefineMask, the second-best method of the baselines, reaching 32.0%, 37.2%, 41.4%, 43.3%, and 44.5% at 5%, 10%, 20%, 30%, and 40% of the labeled data, respectively. Experimental results on this dataset show that our method is still promising.

The three groups of results show that the proposed method effectively utilizes the unlabeled data to improve the accuracy of building instance extraction. In addition, the performance of the method is gradually improved when the ratio of labeled data is increased, suggesting that the proposed model is robust.

To further observe the performance of the proposed method, three images representing challenging scenes from the three datasets are visualized in Fig. 8 with 10% and 30% labeled data as examples. As shown in the first and second columns of the figure, the results of the baseline methods, including Mask R-CNN, RefineMask, SOLOv2, and NB, suffer from missed detection and incomplete buildings, while the proposed method segments the buildings well with precise contours. The visualization results further demonstrate that the proposed method is competitive in extracting building instances.

V. Discussion

A. Ablation Study

To investigate the effectiveness of two key modules, the HDA and CR modules, an ablation study was analyzed on the WHU dataset with 30% labeled and 70% unlabeled RSIs. Specifically, three experiments are conducted as follows: 1) removing HDA (w/o HDA); 2) removing CR (w/o CR); and 3) removing HDA and CR (w/o HDA and CR). The experimental results are reported in Table II.

TABLE II
Ablation Study Results on the WHU Dataset

Method	mAP
Proposed method	**0.658**
w/o HDA	0.645 (1.3% ↓)
w/o CR	0.631 (2.7% ↓)
w/o HDA, CR	0.620 (3.8% ↓)

TABLE III
Comparison of Results With Different Thresholds on Three Datasets

Threshold	mAP		
	WHU	CrowdAI	TCC
0.5	0.653	0.560	0.427
0.7	0.655	0.556	0.422
0.9	**0.658**	0.554	0.416
Adaptive threshold	**0.658**	**0.564**	**0.433**

As shown in the table, removing any module led to a degradation in performance. Specifically, the proposed method exhibits the best performance, achieving the highest mAP of 65.8%. After removing HDA, the mAP decreases by 1.3%. We argue that the HDA module is employed to generate more diversity building features, resulting in a better extracting ability of the teacher model. In addition, after removing CR, the value of mAP is decreased by 2.7% compared with the proposed method. After removing both the HDA and CR modules, the mAP is reduced to 62.0%. These results suggest that both modules help to improve the performance of building instance extraction.

Several examples from the WHU dataset are visualized in Fig. 9 to further observe the effect of the components. In the figure, the differences of the extracted buildings are highlighted in red boxes. Compared with Fig. 9(c), the extracted buildings in Fig. 9(d) and (e) are better. Furthermore, as shown in Fig. 9(f), the proposed method produces clearer contours of building instances. The experimental results show that both the HDA and CR modules contribute to the improvement of the extraction ability.

B. Effect of the Adaptive Threshold Strategy

In Section III-C, an adaptive threshold strategy is used to filter out low-confidence pseudo-labels. To examine the effect of this strategy, three groups of experiments are conducted by setting the threshold values as 0.5, 0.7, and 0.9, respectively. In the experiments, the proposed method takes 30% labeled and 70% unlabeled data of the WHU, CrowAI, and TCC datasets as input, respectively.

As shown in Table III, the best results are obtained when applying the adaptive strategy on all three datasets. We argue that if the fixed threshold is set too low, some false building box labels may be used, leading to more noise and impairing the discriminative ability of the network. When the fixed threshold is set too high, some positive building boxes will be assigned as negative labels, which will hinder the training of the model and decrease the accuracy. The adaptive strategy

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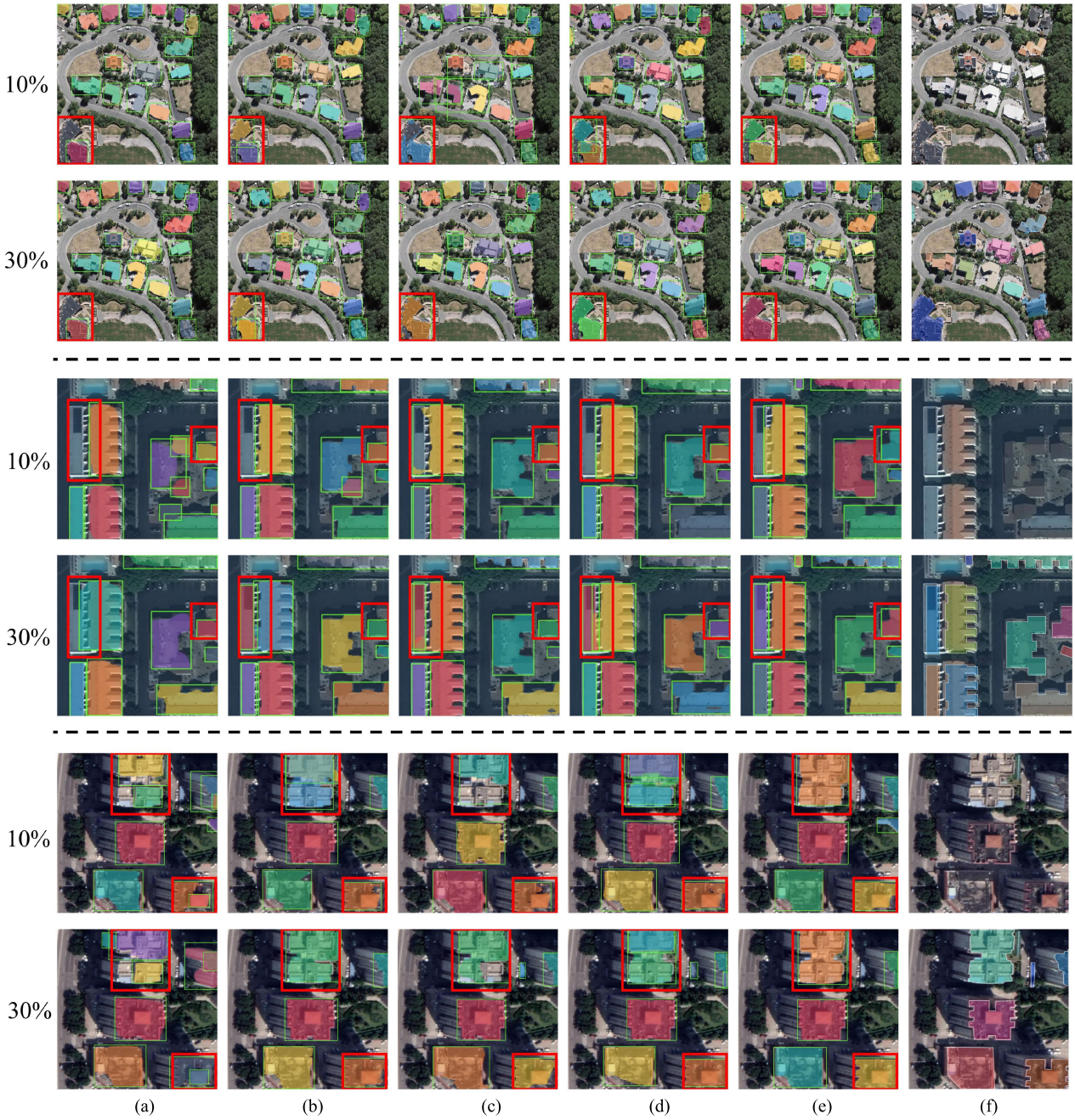


Fig. 8. Visualized results of building instance extraction on three datasets. (a) Mask R-CNN. (b) RefineMask. (c) SOLOv2. (d) NB. (e) Proposed method. (f) Original RSIs/GT.

provides an effective filtering approach with the help of labeled data distributions.

C. Effect of Data Augmentation Strategy

The HDA module consists of a geometric data augmentation strategy and a basic data augmentation strategy. Usually, data augmentations play a critical role in improving model generalization and robustness and is emerging as a major enabler of SSL. To analyze the effect of different data augmentation strategies, the experiments are conducted on the WHU aerial image dataset with 30% labeled and 70% unlabeled data. As shown in Table IV, the mAP on the teacher model decreases by 0.9% and 0.9%, and the mAP on the student model decreases by 0.5% and 0.7% when removing geometric data augmentation and basic data augmentation, respectively. Removing the two strategies decreases the mAP by 1.5% and 1.3% on the teacher model and the student model, respectively. These results demonstrated that both the geometric data augmentation and the basic data augmentation strategies help the learning of the teacher model and improve the performance of the student model.

D. Effect of Edge Detection Operators

The CR module adopts the Sobel operator to further refine building contours. To investigate the effect of the operator, additional experiments are conducted with two substitutable

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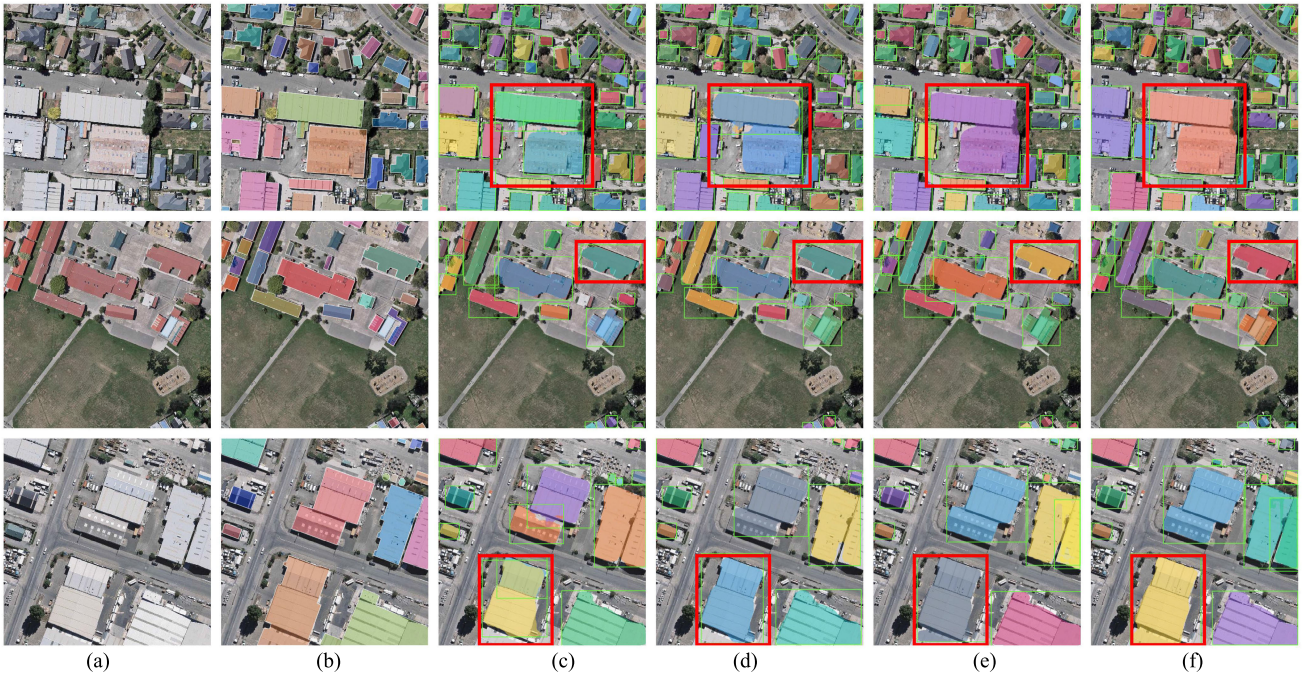


Fig. 9. Illustrative results on WHU to show the effectiveness of HDA and CR. (a) Original RSIs. (b) GT. (c) w/o HDA and CR. (d) w/o CR. (e) w/o HDA. (f) Proposed method.

TABLE IV
COMPARISON OF RESULTS WITH DIFFERENT DATA AUGMENTATION STRATEGIES ON THE WHU DATASET

Augmentation	*mAP*	
	Teacher	Student
HDA	**0.615**	**0.658**
w/o geometric	0.606 (0.9% ↓)	0.653 (0.5% ↓)
w/o basic	0.606 (0.9% ↓)	0.651 (0.7% ↓)
w/o geometric, basic	0.600 (1.5% ↓)	0.645 (1.3% ↓)

operators. Specifically, taking the WHU dataset as an example, the performance of several edge detection operators is discussed, including Sobel [44], Laplace [56], and Canny [57]. As shown in Fig. 10, the best mAP is obtained when applying the Sobel operator. This can be explained by the fact that the Sobel operator contributes to the smoothing of building contours and noise suppression [58].

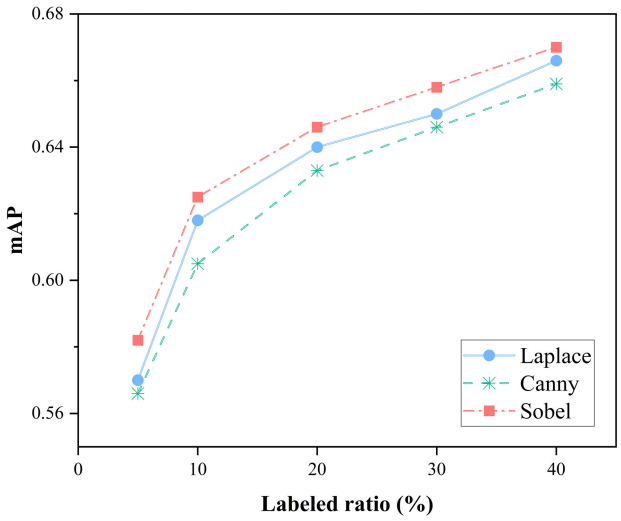


Fig. 10. mAP on several edge detection operators.

E. Experiment With Unlabeled Data Out of Dataset

The above experiments are conducted with both labeled and unlabeled data from a single dataset, where the two types of data have highly consistent building and image features. In this section, seven experiments are conducted to explore the potential of the proposed approach in different data source scenarios. In these experiments, the labeled data are used in the training set of the WHU dataset, and its test set is used for testing, and the unlabeled data are out of the WHU dataset. The first experiment uses the null set as its unlabeled data; that is, no unlabeled data are used for training. The second and third experiments uses the RSIs from CrowdAI and TCC as unlabeled data, respectively. For the remained four experiments, their unlabeled RSIs are sourced from four regions, Tokyo, Shenzhen, Serbia, and New Zealand, respectively. The unlabeled RSIs are downloaded from Google Earth with 500 × 500 pixels at a 0.79-m spatial resolution. Some samples from the four regions are shown in Fig. 11. As shown in the figure, there are large variations in image features across regions. For example, the distribution of buildings varies greatly in the Tokyo in density, and the buildings sparsely distributed in the Serbia region. The experimental results are reported in Table V.

As shown in Table V, combining any of the unlabeled datasets resulted in an improvement in mAP. In particular,

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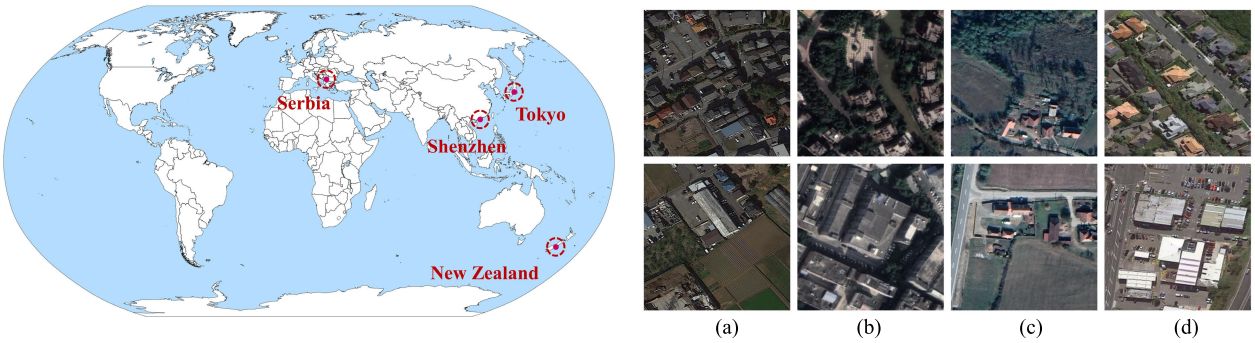


Fig. 11. Distributions and samples of the unlabeled dataset in four regions. (a) Tokyo. (b) Shenzhen. (c) Serbia. (d) New Zealand.

TABLE V
COMPARISON RESULTS WITH UNLABELED DATA OUT OF THE WHU DATASET

Labeled	Unlabeled data source	Unlabeled RSIs	mAP
WHU (4736)	N/A	0	0.635
	CrowdAI	1,200	0.685 (5.0% ↑)
	TCC	1,200	0.677 (4.2% ↑)
	Google Earth-Tokyo	1,200	0.686 (5.1% ↑)
	Google Earth-Shenzhen	1,173	0.682 (4.7% ↑)
	Google Earth-Serbia	1,287	0.672 (3.7% ↑)
	Google Earth-New Zealand	1,224	**0.693** (5.8% ↑)

the RSIs from Serbia resulted in less improvement. This may be due to the fact that there are large forest areas and few building in the region, which is not very effective to learn building features. Similarly, TCC is a typical Chinese building dataset, whose building features are very different from those of WHU, incorporating unlabeled RSIs in TCC provides limited performance improvement. On the other hand, the highest performance was achieved by adding the unlabeled data from New Zealand, which is due to the fact that the WHU dataset originates from New Zealand. This can be explained by the fact that the more similar the unlabeled data features are to the labeled data, the better it is for the learning of the model. In addition, the introduction of RSIs in CrowdAI improves the model performance greatly, probably due to the fact that the RSIs in this dataset originate from four countries, which effectively enriches building features.

F. Model Efficiency

To examine the efficiency of the proposed method, the comparison experiments with four baseline methods are performed on the WHU dataset. Table VI reports the inference time of a single image in milliseconds. Compared with the two fully supervised methods, the inference time of the proposed method is less than that of RefineMask and SOLOv2. Although the inference time of our method is about 6 ms longer than that of Mask R-CNN, it improves the model performance in the limited labeled data scenario. Compared with the semisupervised instance segmentation method, NB, the proposed method achieves an improvement with a small additional computational cost of less than 5%. We argue that the proposed method offers a great trade-off between accuracy and efficiency improvement.

TABLE VI
ABLATION STUDY RESULTS ON THE WHU DATASET

Method	Inf (millisecond)
Mask R-CNN	36.010
RefineMask	51.564
SOLOv2	45.529
NB	40.149
Proposed method	42.632

VI. CONCLUSION

This study proposes a semisupervised method for building instance extraction from HR RSIs. The method designs HDA to support the teacher model training on labeled data and proposes the CR module to refine the extracted contours. The experiments on three datasets show that the proposed method outperforms baselines and is robust. Furthermore, this study shows that the introduction of unlabeled data can enhance the ability to extract building instances. This study provides a new approach for extracting building instances from HR RSI in scenarios with insufficient training samples and a methodological reference for various applications of semisupervised on RSIs.

In the future, the proposed method focuses on improving the quality of pseudo-labels generated from unlabeled data. In addition to the teacher–student framework, more semisupervised techniques, such as consistency regularization, can be integrated to improve the method.

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Fang Fang (Member, IEEE) received the B.S. degree in computer science and technology and the Ph.D. degree in management science and engineering from the China University of Geosciences, Wuhan, China, in 1998 and 2012, respectively.

She is currently an Associate Professor with the School of Computer Science, China University of Geosciences. Her research interests include intelligent interpretation of remote sensing imagery and urban visual intelligence.



Rui Xu received the B.S. degree in software engineering from the China University of Geosciences, Wuhan, China, in 2021, where he is currently pursuing the M.S. degree in software engineering.

His research interests include instance segmentation and semisupervised deep learning for remote sensing.



Shengwen Li (Member, IEEE) received the B.S. degree in computer science and the Ph.D. degree in cartography and geographic information engineering from the China University of Geosciences, Wuhan, China, in 2000 and 2010, respectively.

He is currently an Associate Professor with the School of Computer Science, China University of Geosciences. His research interests include deep learning for remote sensing, spatial–temporal data mining, and knowledge graph.



Qingyi Hao received the B.S. degree in network engineering from the Changsha University of Science and Technology, Changsha, China, in 2021. She is currently pursuing the M.S. degree in software engineering with the China University of Geosciences, Wuhan, China.

Her research interests include real-time semantic segmentation.



Kang Zheng received the B.S. degree in software engineering from the China University of Geosciences, Wuhan, China, in 2021, where he is currently pursuing the M.S. degree in software engineering.

His research interests include semantic segmentation of remote sensing.



Kaishun Wu received the B.S. degree in transportation engineering from Southwest Jiaotong University, Chengdu, China, in 2019, and the M.S. degree in software engineering from the China University of Geosciences, Wuhan, China, in 2022.

His research interests include remote sensing image intelligent interpretation and semisupervised deep learning.



Bo Wan (Member, IEEE) received the B.S. degree in computer science and technology and the Ph.D. degree in cartography and geographic information engineering from the China University of Geosciences, Wuhan, China, in 1998 and 2007, respectively.

He is currently a Professor with the School of Computer Science, China University of Geosciences. His research interests include remote sensing, geographic information system (GIS) modeling, spatial–temporal information mining, and analysis.

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